JP Morgan Industry Project — Summer 2026

Optimal Intraday Delta Hedging

Strategy Framework & Formulations

Problem Setup

The Liq+ book accumulates net delta exposure $\Delta(t)$ continuously throughout the trading day from JPM's options executions. The desk must be **delta-flat overnight**: $\Delta(T) = 0$. The central trade-off is:

Action	Benefit	Cost
Hedge immediately	Minimal delta risk	High transaction / market impact cost
Do not hedge	Minimal cost	Large residual overnight delta risk
Hedge optimally	Balanced	Requires principled optimization rule

Step 1 — The Whalley-Wilmott Band Filter

Overview

At every point in time, the system monitors the real-time net delta $\Delta(t)$ of the entire options book and compares it against dynamic no-trade boundaries $[\Delta^-(t), \Delta^+(t)]$:

- **Scenario A (Inside bands):** $\Delta^-(t) \leq \Delta(t) \leq \Delta^+(t)$. The system does nothing. No spreads crossed, no market impact, zero transaction cost.
- **Scenario B (Band breached):** $\Delta(t)$ crosses $\Delta^+(t)$ or $\Delta^-(t)$. The execution engine is triggered. The system computes the required hedge quantity and passes it to Step 2.

Whalley-Wilmott Band Formula

The Whalley-Wilmott (1997) framework derives the optimal no-trade band analytically under proportional transaction costs. The half-width of the band around the Black-Scholes delta is:

$$H(t) = \left(\frac{3}{2} \cdot \frac{c \cdot \Gamma(t)^2 \cdot \sigma(t)^2}{\lambda(t)} \right)^{1/3} \tag{1}$$

where the no-trade region is $\left[\Delta^{\mathrm{BS}}(t) - H(t),\ \Delta^{\mathrm{BS}}(t) + H(t)\right]$.

Symbol	Description	Source
c	Proportional transaction cost (half-spread)	Tick+
$\Gamma(t)$	Portfolio gamma (second derivative of value w.r.t. S)	MI5 / real-time greeks
$\sigma(t)$	Realized spot volatility of underlying	Tick+
$\lambda(t)$	Time-varying risk aversion (see below)	Calibrated
$\Delta^{\mathrm{BS}}(t)$	Black-Scholes hedge delta of the book	MI5

Time-Varying Risk Aversion $\lambda(t)$

A key extension over standard Whalley-Wilmott is to allow $\lambda(t)$ to increase as market close approaches. The desk faces a *hard* overnight constraint $\Delta(T) = 0$, so risk tolerance shrinks to zero at $t = T$.

$$\lambda(t) = \lambda_0 + \frac{\lambda_\infty - \lambda_0}{\left(1 - \frac{t}{T}\right)^{\gamma}}, \qquad t \in [0, T) \tag{2}$$

$\lambda(t)$ plays a **dual role**:

1. **Trigger:** Via equation (1), higher $\lambda(t) \Rightarrow$ narrower band $H(t) \Rightarrow$ more frequent hedging triggers near close.
2. **Execution:** Higher $\lambda(t) \Rightarrow$ more aggressive Almgren-Chriss schedule after each trigger (see Step 2).

> **Intuition:** At 10:00am the bands are wide — small delta drifts are tolerated. At 3:45pm the bands have collapsed to near-zero — almost any deviation triggers immediate execution. The EOD constraint $\Delta(T) = 0$ emerges naturally from $\lambda(t) \to \infty$ as $t \to T$, rather than being imposed as a separate hard constraint.

Step 2 — Execution Engine (Almgren-Chriss)

The Almgren-Chriss Loss Function

Once a band breach triggers execution, the system must liquidate/acquire $\Delta(t)$ shares optimally. The Almgren-Chriss (2001) mean-variance framework minimizes:

$$\mathcal{L} = \underbrace{E\left[\int_t^T \eta(s)\, v(s)^2\, ds\right]}_{\text{execution cost}} + \lambda(t)\, \underbrace{E\left[\int_t^T \sigma(s)^2\, \tilde{\Delta}(s)^2\, ds\right]}_{\text{delta risk cost}} \tag{3}$$

Symbol	Description	Source
$v(s) = -\dot{\tilde{\Delta}}(s)$	Trading velocity (shares/sec)	Control variable
$\tilde{\Delta}(s)$	Residual delta to be executed	From band breach
$\eta(s)$	Temporary market impact coefficient	Calibrated from Tick+
$\sigma(s)$	Spot volatility	Tick+
$\lambda(t)$	Risk aversion at trigger time t	From eq. (2)

Almgren-Chriss Closed-Form Optimal Trajectory

For a residual inventory $\tilde{\Delta}_0$ at trigger time t, the closed-form optimal execution trajectory is:

$$\tilde{\Delta}^*(s) = \tilde{\Delta}_0 \cdot \frac{\sinh(\kappa\,(T-s))}{\sinh(\kappa\,(T-t))}, \qquad \kappa = \sqrt{\frac{\lambda(t)\,\sigma^2}{\eta}} \tag{4}$$

The optimal trading rate is:

$$v^*(s) = \tilde{\Delta}_0 \cdot \frac{\kappa\,\cosh(\kappa\,(T-s))}{\sinh(\kappa\,(T-t))} \tag{5}$$

where κ controls urgency: large κ (from large $\lambda(t)$ near close) front-loads execution; small κ spreads it evenly.

Strategies

Strategy 1 — Dynamic Re-optimization (Myopic Baseline)

Mechanism: Every time the W-W band is breached, re-solve the Almgren-Chriss problem from scratch using current $\tilde{\Delta}(t)$ and remaining time $T - t$. Discard any prior execution schedule.

1. Band breach detected: residual delta $= \tilde{\Delta}(t)$.
2. Compute $\kappa = \sqrt{\lambda(t)\sigma^2/\eta}$ using current $\lambda(t)$ from eq. (2).
3. Execute closed-form trajectory from eq. (4).
4. On next breach, repeat from step 1.

> **Key assumption:** No more delta arrivals expected after this breach. Purely reactive — no predictive model. This is the *smart practitioner baseline*: it uses optimal execution but ignores the statistical structure of future arrivals.

Strategy 2 — Arrival-Conditioned Execution

Mechanism: Same as Strategy 1, but the execution speed is conditioned on a forward-looking model of future delta arrivals. Before executing, the system asks: *how much more delta is likely to arrive before close?*

Stochastic Delta Arrivals Formulation

The delta book evolves as:

$$d\Delta(t) = -v(t)\,dt + dN(t) \tag{6}$$

Symbol	Description
$v(t)$	Hedging trades (control variable)
$dN(t)$	New delta arriving from JPM's options executions (stochastic)
$\Delta(T) = 0$	Hard EOD constraint

Hawkes Process for Arrival Intensity

$dN(t)$ is modeled as a **Hawkes process** — a self-exciting point process that captures the empirical clustering of options executions:

$$\mu_N(t) = \mu_0 + \sum_{t_i < t} \alpha\, e^{-\beta(t-t_i)} \tag{7}$$

where μ_0 is the baseline arrival rate, α is the self-excitation amplitude, β is the decay rate, and the sum runs over all past arrival times t_i.

Conditioned Execution Rule

At trigger time t, compute the expected future delta load:

$$\hat{N}(t) = E\left[\int_t^T dN(s) \,\middle|\, \mathcal{F}_t\right] \tag{8}$$

- If $\hat{N}(t)$ is large (burst expected): **slow down** — the arriving delta may offset current inventory, wasting execution cost.
- If $\hat{N}(t) \approx 0$ (book near final): **speed up** — execute aggressively, current inventory is close to EOD inventory.

> **Key distinction from Strategy 1:** dN(t) in the Hawkes process models JPM's *options execution flow* (the source of delta), not the W-W band breaches themselves. Band breaches are downstream consequences of the arrival process.

Strategy 3 — Intraday Urgency Scheduling

Mechanism: Simpler than Strategy 2 — no full arrival model. $\lambda(t)$ alone drives execution urgency. Calibrated directly from historical Liq+ data.

- Uses the same $\lambda(t)$ from eq. (2) both for the W-W trigger and the A-C execution schedule.
- No Hawkes process. No conditional forecasting.
- Calibrated by asking: *when did most delta historically arrive? When was residual EOD risk highest?*
- Computationally cheap. If performance is close to Strategy 2, the Hawkes process complexity is not justified in production.

> **Research value:** If Strategy 3 $\approx$ Strategy 2 in backtest, the finding is: a simple urgency ramp captures most of the gain from the full stochastic arrival model. That is a practically important and publishable result.

Benchmarks

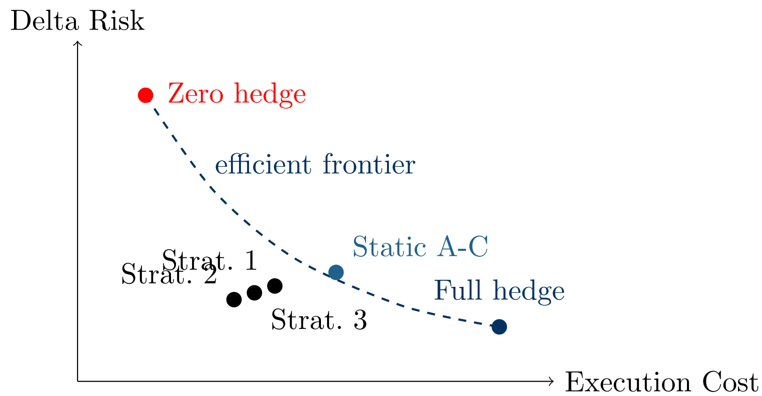
All three strategies are evaluated against the following benchmarks:

Benchmark	Description	Purpose
Static Almgren-Chriss	Solve A-C once at first band breach. Never re-solve even as new delta arrives throughout the day.	Best static schedule; real bar for Strategies 1–3.
Full Hedge	Execute every delta arrival immediately at market price.	Upper bound on cost, lower bound on risk.
Zero Hedge	Do nothing all day; unwind everything at close.	Lower bound on cost, upper bound on risk.

The primary evaluation metric is improvement in the combined loss $\mathcal{L}$ (eq. (3)) over **Static Almgren-Chriss**. Full hedge and zero hedge frame the extreme endpoints of the cost-risk frontier but are not the meaningful bar.

Efficient Frontier Visualization

The cleanest evaluation is an **efficient frontier** in (execution cost, delta risk) space:



Each strategy should push the frontier **down-left** relative to Static Almgren-Chriss, reducing both cost and risk simultaneously.

Summary of Framework

Component	Framework	Answers	New Ingredient
Trigger	Whalley-Wilmott + $\lambda(t)$	When & how much to hedge	—
Strategy 1	Rolling A-C (myopic)	How to execute	React to arrivals
Strategy 2	A-C + Hawkes process	How to execute	Predict arrivals
Strategy 3	A-C + $\lambda(t)$ ramp only	How to execute	Time-of-day urgency